



NEWSLETTER

AUGUST 2024

AERODYNAMICS

THE CHALLENGES OF
MAKING THE PERFECT
BODYWORK

THE PACKAGING NIGHTMARE

LEARN HOW WE FIT A
GIANT BATTERY IN A
SMALL CAR



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WHAT IS AUS RACING

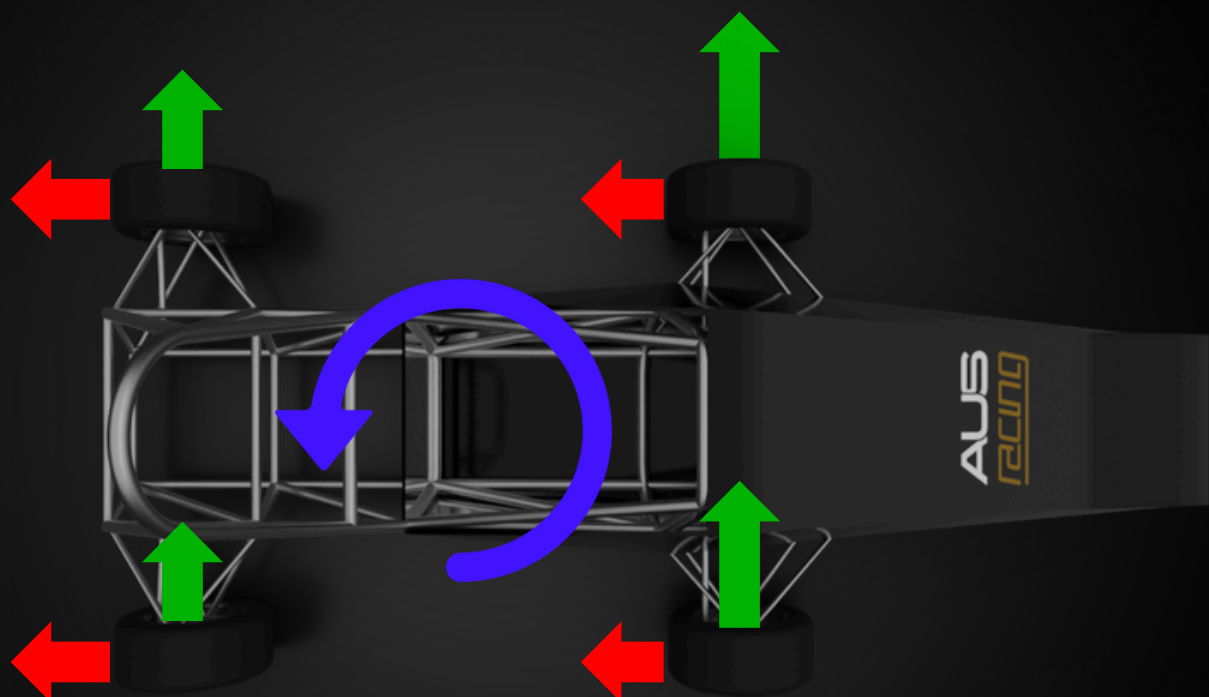
AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

MAKING IT TURN



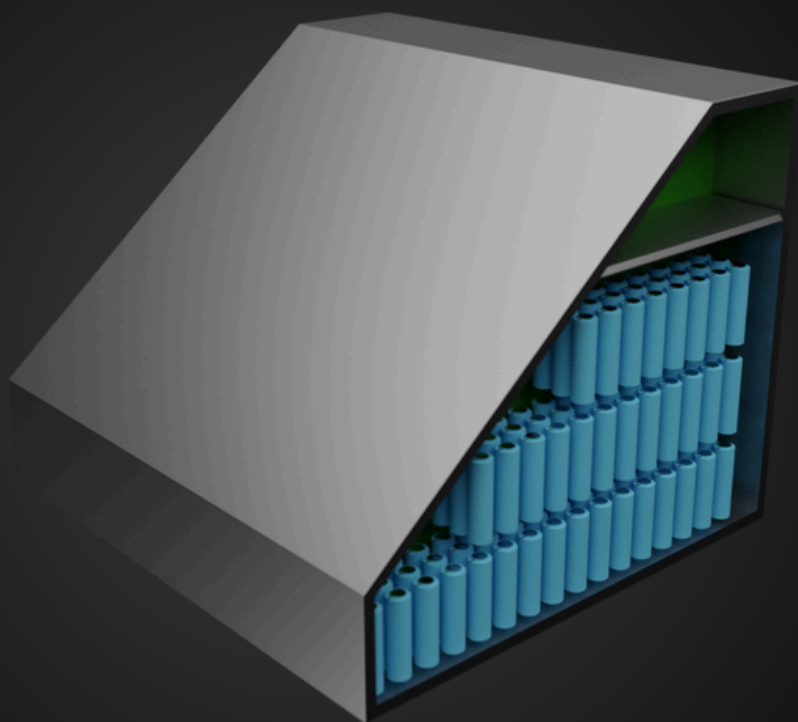
Getting the car to turn the way the driver wants is a challenge left to vehicle dynamics engineers. While entering a corner the car needs to be easier to make a sharp turn (oversteer) and harder on exit (understeer).

In order to achieve such characteristics, the vehicle dynamics are tuned to perfection

through the braking system, suspension, steering, weight distribution, aerodynamics and most importantly, the tires.

All these factors influence the yaw moment on the car. Yaw moment is the force that causes the car to rotate. A higher yaw moment will lead to the car turning faster.

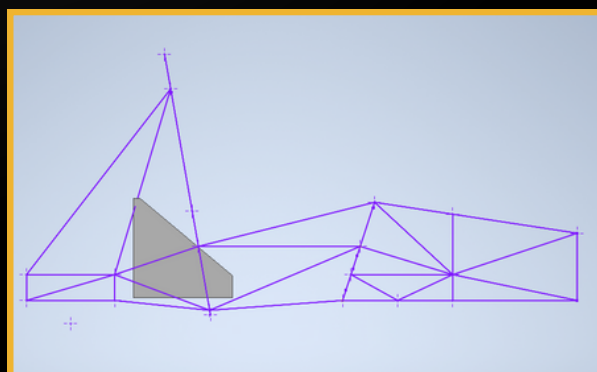
SIZING UP THE BATTERY



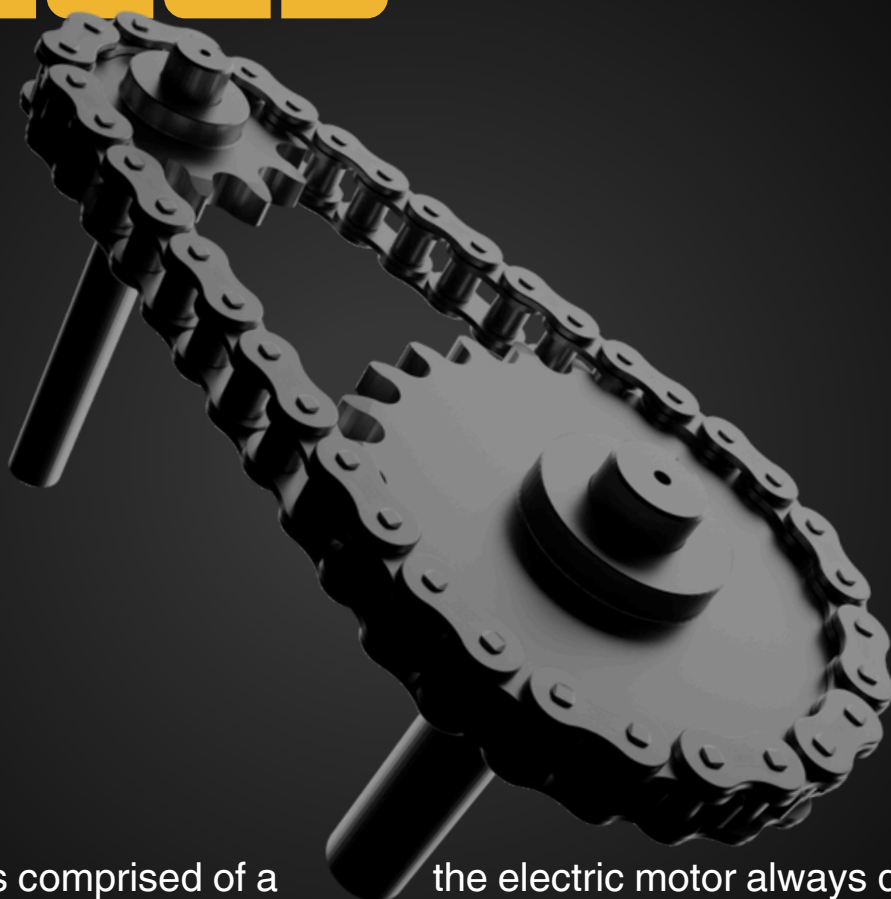
The battery container is meticulously designed to safely hold all 648 cells and other electronics without compromising safety, performance and space.

The small size of the car makes it extremely important to optimize the volumetric efficiency of the battery. Volumetric efficiency refers to how effectively the space in the battery is used to reduce the size as much as possible.

Our cells are organized in a way to offer a 40° angle such that it can be placed as close to the reclined driver's seat as possible to efficiently use the area under the driver's back.



MOTOR TO WHEELS



Our drivetrain is comprised of a simple system, however, it is a very intricate system. It involves extensive research and development to design an effective drivetrain with minimum energy losses and maximizing reliability.

Our drivetrain comprises of a simple chain drive to ensure the the speed and torque at the wheels is exactly what we need. A transmission is not required as

the electric motor always delivers the ideal power.

Getting the torque from the chain to the wheel is a challenge in itself. Unfortunately, the wheel is never at the same level relative to the chassis because of the suspension system.

To circumvent this, tripod joints are used. They are a special kind of joint that allow up to 12° of motion in 2 axes.

CONTROLLING AIRFLOW



The aerodynamics of the car is among the most challenging parts of the car. It requires a deep understanding of fluid mechanics to understand the airflow around the car and run simulations accordingly.

The bodywork is designed such that the car is as aerodynamic as possible while pushing air into the undertray to optimize it's

performance.

The undertray of the car acts similar to a suction cup. It creates a low pressure zone below the car so as to push the car down by what is known as downforce.

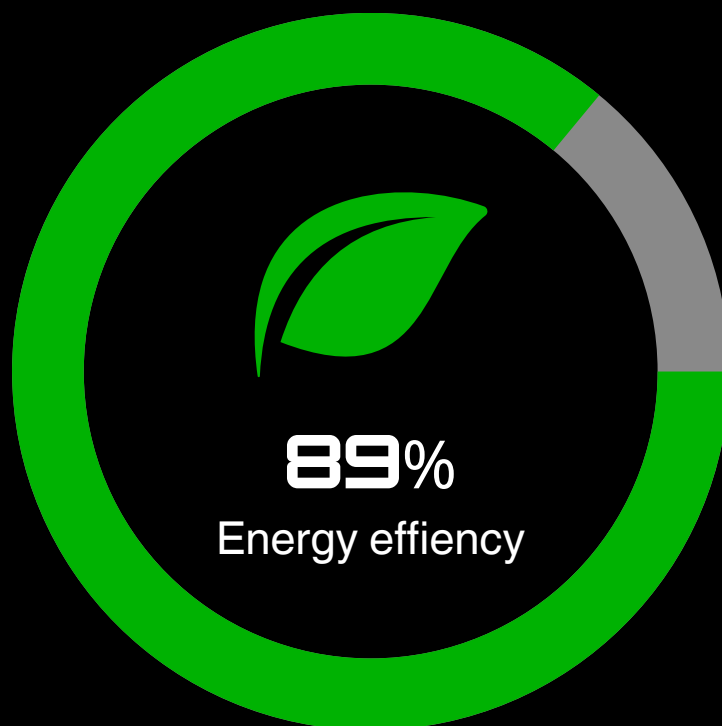
The bodywork also features a diffuser which is a necessary component to make the undertray as effective as possible.

OUR GREEN INITIATIVE

Formula Student provides AUS students with numerous opportunities, from serving as a platform to demonstrate our technical expertise to granting us access to a vast network of companies, institutions, and talented peers. Yet, these advantages come at a significant cost. We have observed that such advancements often have adverse environmental effects, particularly as carbon emissions frequently go unaddressed. For instance, the carbon emissions of internal combustion engines (ICE) used at such a competition can reach up to 8 kg of CO₂ per race. These estimates do not even account for logistical factors, such as transportation, which can add as much as 100 kg of CO₂ per trip.



The AUS community strongly commits to environmental stewardship and prioritizes sustainability over mere growth. As a team, we share this dedication and recognize the importance of integrating environmental considerations from the outset of our project. In light of this, we are committed to maintaining transparency regarding our CO2 emissions and factoring them into our technical decisions. For instance, we have chosen to utilize an electric vehicle (EV) motor, which can have life cycle emissions as low as 50 grams of CO2 equivalent (tCO2e), significantly lower than those of ICE. Additionally, EVs boast an impressive energy efficiency of 89%, far surpassing that of ICE with an efficiency of just 20%.



We shall continue to contribute our unstaggering commitment to ecological prosperity can help bring bigger goals of sustainability into fruition. We shall continue to contribute to this cause in our own authentic way and to the best of our capabilities. For now, we are expanding our scope of incorporating sustainability and environmental conscience into our initiative, but we hope these small steps turn into bigger ones.

MEET OUR NEW SPONSOR

We are excited to announce our latest sponsor, Ansys. They have generously provided us with industry standard software for the development of our car.

Ansys is a collection of engineering software used for analyzing the behavior of structures, fluids etc. It enables engineers to predict performance and optimize designs by solving complex problems through multifaceted simulations.

Ansys is known for its commitment to innovation, regularly updating its software to include the latest advancements in simulation technology, artificial intelligence, and cloud computing. The company serves a diverse customer base, ranging from small startups to large multinational corporations, and is deeply involved in various academic and research initiatives to advance the field of engineering simulation.

We thank them for their kind support and look forward to more milestones!



[ansys.com](https://www.ansys.com)

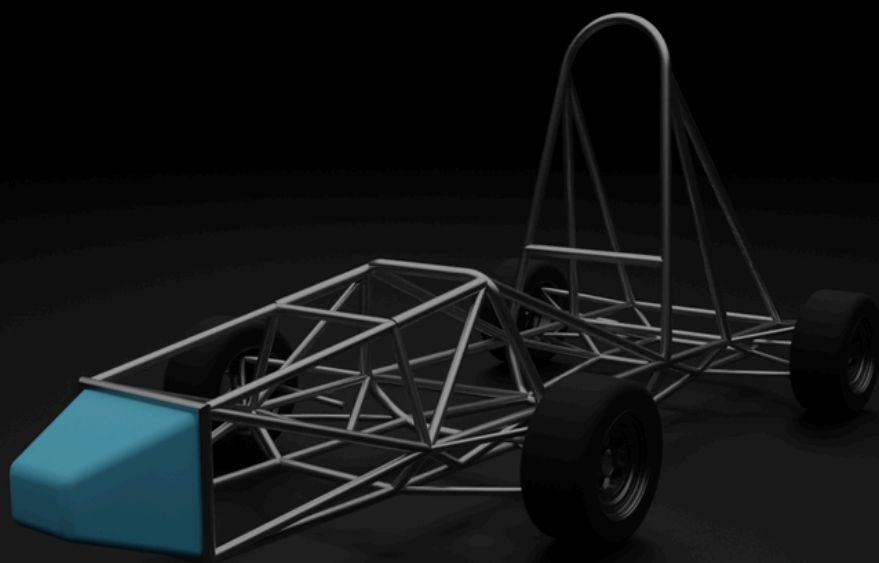
HOW YOU CAN SPONSOR US

What if you were at the forefront of engineering innovation?

What if your company was responsible for bringing our aspiring engineers' ideas to life?

What if your business could get its message across the world through Formula Student?

If your answer to any of these questions is yes, click [here](#) and check out what we have in store for our generous sponsors and learn how to become one yourself!



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